Session 9: Re-entry Dynamics and Mission End-of-Life Planning

Task 1: Explain why peak deceleration is independent of the ballistic coefficient, and why re-entry vehicles are blunt rather than pointed

1. Why is peak deceleration independent of the ballistic coefficient

- **Ballistic Coefficient Definition:** $B=\frac{m}{C_d A}$, where m is mass, C_d is he drag coefficient, and A is reference area.
- **Velocity Profile:** Under the Allen-Eggers model for planar ballistic entry, velocity as a function of altitude h is.

$$v(h)=v_E \exp\left(\frac{-\rho_0}{2B\gamma \sin\gamma_E} e^{-\gamma h}\right)$$

- **Peak Deceleration Derivation:** Taking the time derivative of velocity to find deceleration $a(h)=\frac{-dv}{dt}=-v\frac{dv}{dh}\sin\gamma_E$ and maximizing it yields:

$$a_{\max}=\frac{v_E^2 \cdot \gamma \cdot \sin\gamma_E}{2e}$$

- **Conclusion:** The ballistic coefficient B cancels out of the equation for peak deceleration $a_{\max}$. Varying B changes the **altitude** where peak deceleration occurs ($h_{peak-g}=\frac{1}{\gamma}\ln\left[\frac{\rho_0}{B\gamma\sin\gamma_E}\right]$), but **does not change the maximum g-load experienced**.
-

2. Why are re-entry vehicles blunt rather than pointed:

- **Stagnation Point Heat Flux Relation:** According to Fay-Riddell heat transfer theory, stagnation convective heating rate $\dot{q}_s$ scales inversely with the square root of the nose radius (R_n):

$$\dot{q}_s \propto \frac{1}{\sqrt{R_n}}$$

- **Shock Wave Mechanics:** A blunt nose shape creates a strong detached bow shock wave in front of the vehicle. This pushes the ultra-high temperature

shock layer away from the vehicle skin, dumping over 90% of the frictional heat energy into the surrounding airstream rather than absorbing it into the structure. A pointed nose produces an attached shock, causing extreme localized aerodynamic heating that melts the tip.

Task 2: Peak Deceleration

Parameters:

- Entry velocity: $v_E = 7.8 \text{ km/s} = 7800 \text{ m/s}$
- Reciprocal atmospheric scale height: $\gamma \approx \frac{1}{7.2 \text{ km}} = 1.389 \times 10^{-4} \text{ m}^{-1}$
- Sea-level gravity: $g_0 = 9.81 \text{ m/s}^2$
- Euler's number: $e \approx 2.71828$

Allen-Eggers Formula:

$$a_{\max} = \frac{v_E^2 \cdot \gamma \cdot \sin \gamma_E}{2e \cdot g_0} [\text{in } g\text{'s}]$$

Calculations:

1. For $\gamma_E = 2^\circ$ **(**$\sin 2^\circ \approx 0.03490$**):**

$$a_{\max} = \frac{(7800)^2 \cdot (1.389 \times 10^{-4}) \cdot 0.03490}{2 \cdot 2.71828 \cdot 9.81} \approx 5.3 \text{ g} (52 \text{ m/s}^2)$$

2. For $\gamma_E = 5^\circ$ **(**$\sin 5^\circ \approx 0.08716$**):**

$$a_{\max} = \frac{(7800)^2 \cdot (1.389 \times 10^{-4}) \cdot 0.08716}{2 \cdot 2.71828 \cdot 9.81} \approx 13.8 \text{ g} (135.4 \text{ m/s}^2)$$

3. For $\gamma_E = 10^\circ$ **(**$\sin 10^\circ \approx 0.17365$**):**

$$a_{\max} = \frac{(7800)^2 \cdot (1.389 \times 10^{-4}) \cdot 0.17365}{2 \cdot 2.71828 \cdot 9.81} \approx 27.5 \text{ g} (269.7 \text{ m/s}^2)$$

Human Survivability Evaluation:

- $\gamma_E = 2^\circ$ **(**≈ 5.3 g**):** Fully survivable and comfortable for human crew in a supine position (standard lifting/guided entry corridor).
- $\gamma_E = 5^\circ$ (≈ 13.8 g): Tolerable only during emergency aborts (e.g., ballistic descent mode). Causes severe chest compression and vision degradation; near human physiological tolerance limit for short durations.
- $\gamma_E = 10^\circ$ (≈ 27.5 g): Unsurvivable for human astronauts; causes severe internal injury/blackout. Acceptable only for uncrewed cargo capsules, sample return vehicles, or warheads.

Task 3: Entry Profile

Capsule Parameters:

- Entry Velocity: $v_e = 7800\ \text{m/s} (7.8\ \text{km/s})$
- Atmospheric Scale Height: $H = 7500\ \text{m} (7.5\ \text{km})$
- Sea-Level Air Density: $\rho_0 = 1.225\ \text{kg/m}^3$
- Ballistic Coefficient: $B = 300\ \text{kg/m}^2$
- Flight-path angle: $\gamma_E = 6^\circ$
- Vehicle Nose Radius $R_n = 1.0\ \text{m}$

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
# Entry & Atmospheric Parameters
ve = 7800.0 # Entry velocity (m/s)
H = 7500.0 # Scale height (m)
rho0 = 1.225 # Sea-level density (kg/m^3)
beta = 300.0 # Ballistic coefficient (kg/m^2)
gamma = np.radians(6) # Flight-path angle (rad)
```

```
# Altitude vector from 120 km down to 0 km
h = np.linspace(120e3, 0, 1000)
# Exponential atmospheric density model
rho = rho0 * np.exp(-h / H)
# Allen-Eggers velocity profile
v = ve * np.exp(-(rho0 * H) / (2 * beta * np.sin(gamma)) * np.exp(-h / H))
# Deceleration (g-load)
a_g = (rho * v**2 / (2 * beta)) / 9.81
# Stagnation heating rate (W/cm^2) for Rn = 1.0 m
q_dot = (1.7415e-4 * np.sqrt(rho / 1.0) * v**3) / 1e4
# Peak values and altitudes
idx_peak_a = np.argmax(a_g)
h_peak_a = h[idx_peak_a] / 1000.0
max_a = a_g[idx_peak_a]
idx_peak_q = np.argmax(q_dot)
h_peak_q = h[idx_peak_q] / 1000.0
max_q = q_dot[idx_peak_q]
print(f"Peak Deceleration: {max_a:.2f} g at {h_peak_a:.2f} km altitude")
print(f"Peak Stagnation Heating: {max_q:.2f} W/cm^2 at {h_peak_q:.2f} km altitude")
# Plotting the Entry Profile
fig, ax1 = plt.subplots(figsize=(8, 6))
color = 'tab:red'
ax1.set_xlabel('Deceleration (g)', color=color)
ax1.plot(a_g, h/1000, color=color, linewidth=2, label='Deceleration (g)')
ax1.tick_params(axis='x', labelcolor=color)
ax1.set_ylabel('Altitude (km)')
ax1.set_ylim(0, 90)
ax1.set_xlim(0, 18)
ax1.grid(True, linestyle='--', alpha=0.5)
ax2 = ax1.twiny()
color = 'tab:orange'
ax2.set_xlabel('Stagnation Heating (W/cm²)', color=color)
ax2.plot(q_dot, h/1000, color=color, linewidth=2, linestyle='--', label='Stagnation Heating')
ax2.tick_params(axis='x', labelcolor=color)
plt.title('Entry Profile: Deceleration and Heating vs Altitude\n($\\beta = 300\{ kg/m}^2, \\gamma = 6^\\circ$)')
plt.show()
```

Task 4: Deorbit Budget

Parameters:

- Earth Radius: $R_E = 6378 \text{ km}$
- Earth Gravitational Parameter: $\mu = 398600 \text{ km}^3/\text{s}^2$

- Target Perigee Altitude: $h_p = 50 \text{ km} \Longrightarrow r_p = 6378 + 50 = 6428 \text{ km}$

1. Deorbit from 600 km **Orbit (**$r_1 = 6978 \text{ km}$**):**

- Circular orbit velocity at 600 km:

$$v_{c1} = \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{398600}{6978}} \approx 7.558 \text{ km/s}$$

- Semi-major axis of transfer ellipse ($r_a = 6978 \text{ km}, r_p = 6428 \text{ km}$):

$$a_t = \frac{r_1 + r_p}{2} = \frac{6978 + 6428}{2} = 6703 \text{ km}$$

- Apogee velocity on transfer orbit:

$$v_{a1} = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_t}\right)} = \sqrt{398600\left(\frac{2}{6978} - \frac{1}{6703}\right)} \approx 7.401 \text{ km/s}$$

- **Required Retrograde** Δv_{600}**:**

$$\Delta v = v_{c1} - v_{a1} = 7.558 - 7.401 = 0.157 \text{ km/s} = 157 \text{ m/s}$$

2. Comparison with 400 km **Orbit Case (**$r_1 = 6778 \text{ km}$**):**

- Circular orbit velocity at 400 km:

$$v_{c1} = \sqrt{\frac{398600}{6778}} \approx 7.669 \text{ km/s}$$

- Semi-major axis ($r_a = 6778 \text{ km}, r_p = 6428 \text{ km}$):

$$a_t = \frac{6778 + 6428}{2} = 6603 \text{ km}$$

- Apogee velocity on transfer orbit:

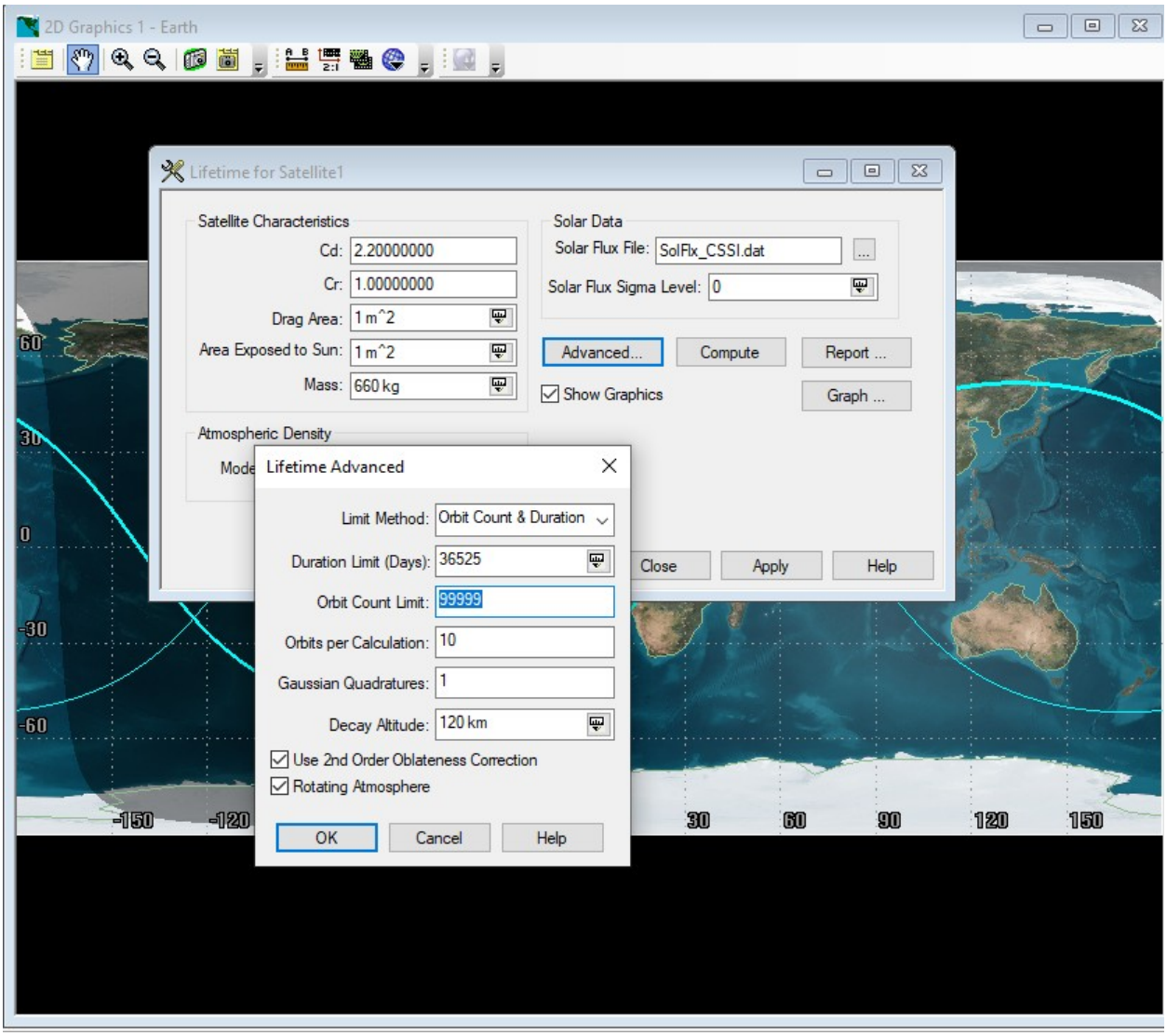
$$v_{a1} = \sqrt{398600\left(\frac{2}{6778} - \frac{1}{6603}\right)} \approx 7.566 \text{ km/s}$$

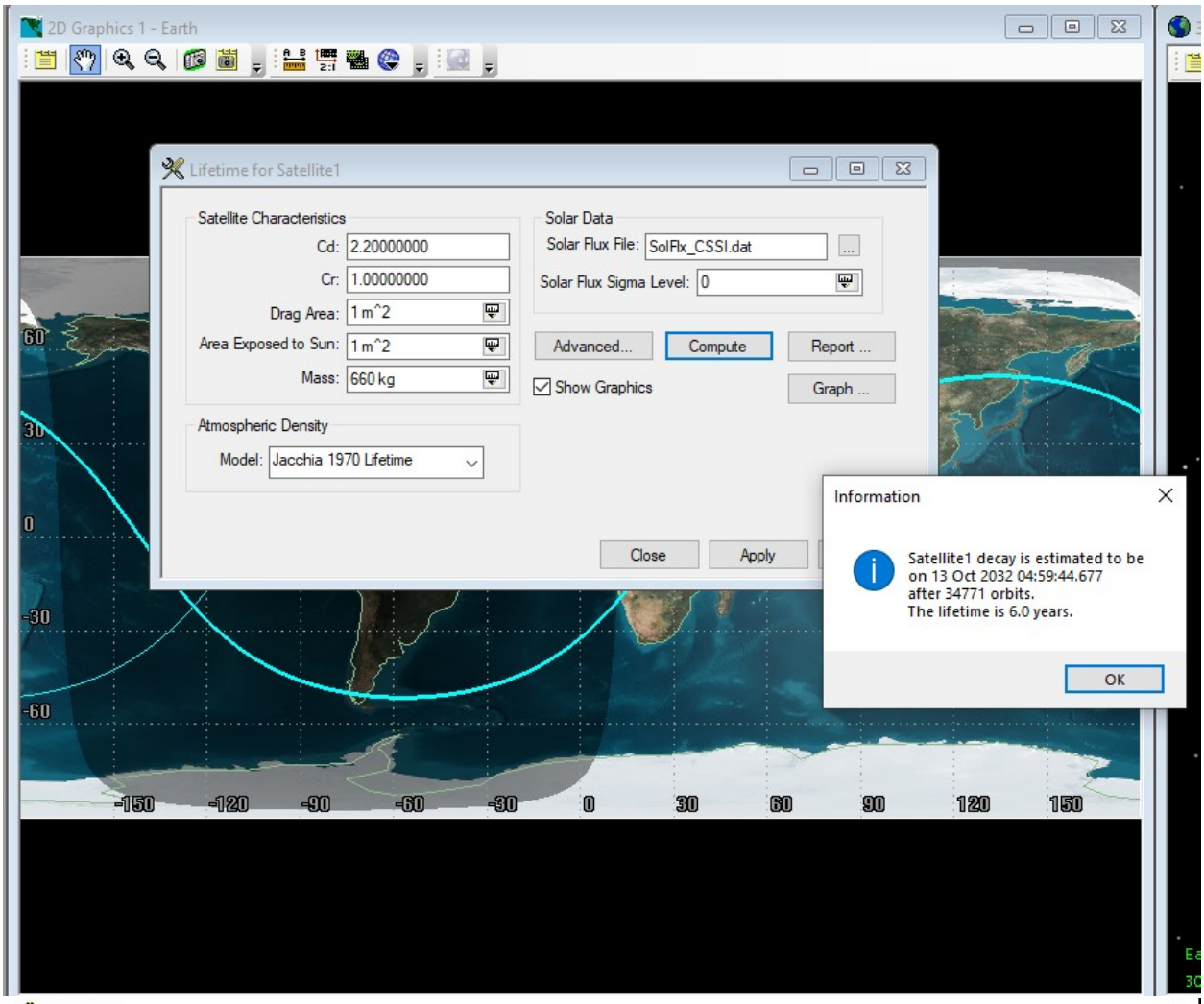
- **Required Retrograde Δv_{400}:**

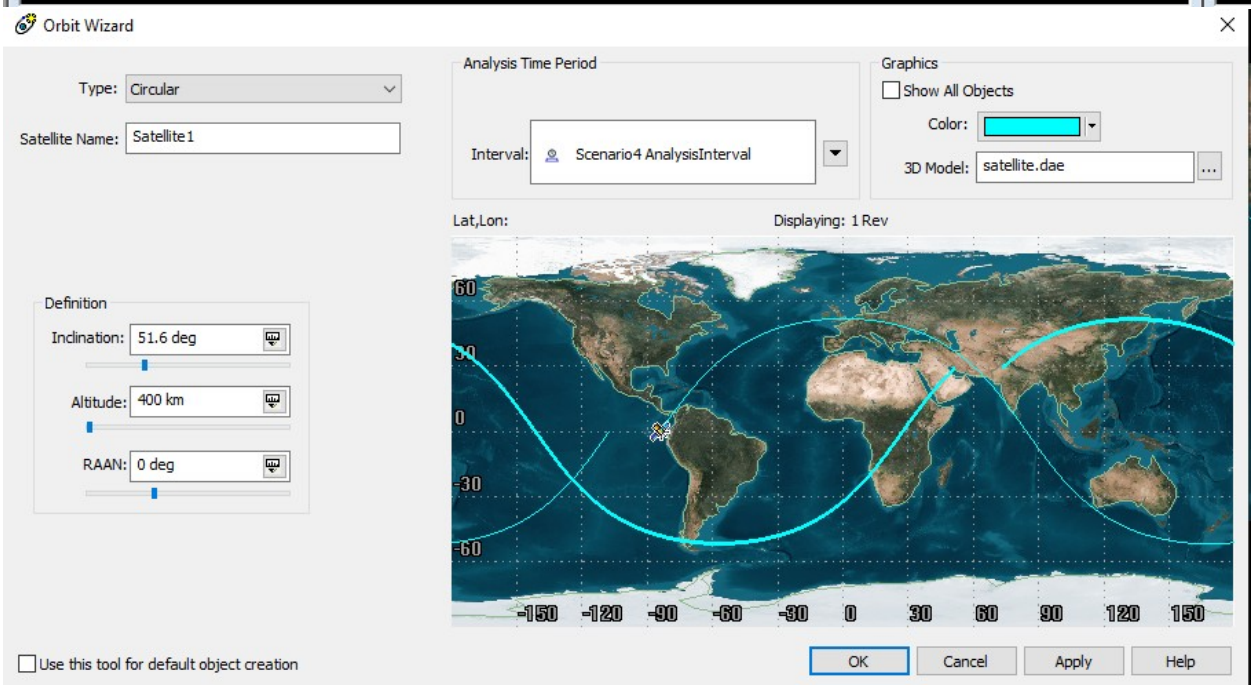
$$\Delta v = 7.669 - 7.566 = 0.103 \text{ km/s} = 103 \text{ m/s}$$

- **Summary Comparison:** Deorbiting from 600 km requires ≈ 157 m/s, which is significantly higher than the ≈ 103 m/s required from 400 km due to the larger orbit drop required ($\Delta h = 550$ km vs 350 km).

Task 5: STK


2D Graphics 1 - Earth
Lifetime for Satellite1
Satellite Characteristics
Cd: 2.20000000
Cr: 1.00000000
Drag Area: 1 m^2
Area Exposed to Sun: 1 m^2
Mass: 660 kg
Atmospheric Density
Mode
Solar Data
Solar Flux File: SolFlx_CSSI.dat
Solar Flux Sigma Level: 0
Advanced...
Compute
Report ...
Show Graphics
Graph ...
Close
Apply
Help
Lifetime Advanced
Limit Method: Orbit Count & Duration
Duration Limit (Days): 36525
Orbit Count Limit: 99999
Orbits per Calculation: 10
Gaussian Quadratures: 1
Decay Altitude: 120 km
Use 2nd Order Oblateness Correction
Rotating Atmosphere
OK
Cancel
Help
60
30
0
-30
-60
-150
-120
30
60
90
120
150


2D Graphics 1 - Earth
Lifetime for Satellite1
Satellite Characteristics
Cd: 2.20000000
Cr: 1.00000000
Drag Area: 1 m^2
Area Exposed to Sun: 1 m^2
Mass: 660 kg
Atmospheric Density
Model: Jacchia 1970 Lifetime
Solar Data
Solar Flux File: SolFlx_CSSI.dat
Solar Flux Sigma Level: 0
Advanced...
Compute
Report ...
Show Graphics
Graph ...
Close
Apply
Information
Satellite1 decay is estimated to be
on 13 Oct 2032 04:59:44.677
after 34771 orbits.
The lifetime is 6.0 years.
OK
60
30
0
-30
-60
-150
-120
-90
-60
-30
0
30
60
90
120
150


Orbit Wizard
Type: Circular
Satellite Name: Satellite1
Analysis Time Period
Interval: Scenario4 AnalysisInterval
Graphics
Show All Objects
Color:
3D Model: satellite.dae
Lat,Lon:
Displaying: 1 Rev
Definition
Inclination: 51.6 deg
Altitude: 400 km
RAAN: 0 deg
Use this tool for default object creation
OK
Cancel
Apply
Help
60
30
0
-30
-60
-150
-120
-90
-60
-30
0
30
60
90
120
150